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Original Article

Effect of Exergaming on Physiological Response and Enjoyment During Recess in Elementary School-Aged Children: A Pilot Study

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Abstract

Objective: With an increasing rate of childhood obesity, it is critical to find fun and effective ways of providing children with opportunities to be physically active while at school to mediate this outcome. Using games as a mode of exercise (Exergaming) could be one such way. We aimed to compare the physiologic response and enjoyment of elementary school children when playing Xbox® Kinect® versus participating in regular activities during recess.
Materials and Methods: Twenty-four children participated individually once in Kinect and once in regular activity sessions on separate days during recess for the allotted time. Sessions were at least 1 week apart, and the order of the sessions was randomized. During the Kinect session, participants played the Reflex Ridge game. During the regular session, participants engaged in self-selected activities. Energy expenditure, metabolic equivalents of task (METS), time in moderate to vigorous activity, percentage of activity in sedentary, light, moderate, vigorous and very vigorous levels, heart rate, rate of perceived exertion (RPE), and Physical Activity Enjoyment Scale were measured for each session and compared using a paired samples *t*-test.
Results: Significantly greater RPE ($P = 0.024$), percentage of activity at sedentary ($P = 0.019$), very vigorous ($P < 0.001$) levels, and lesser percentage of activity at moderate level ($P = 0.008$) were noted during the Kinect session.
Conclusion: Playing Kinect was comparable to playing outside with classmates. When necessary, such as in inclement weather, whole-body interactive videogames can be a viable alternative physical activity. However, this has to be validated with a larger sample study.

Introduction

The rapid increase in the prevalence of overweight (OW) and obesity among children has been acknowledged as an important public health concern.[1] A number of factors have been identified that contribute to this issue, including socioeconomic status, environment, eating behaviors, time spent in sedentary pursuits, and physical activity. With the release of the national physical activity guidelines in 2008, emphasis on ways to increase participation in physical activity among children has been gaining attention; however, physical activity levels among youth remain low.[2] Troiano et al. reported that only 42% of children aged 6–11 years meet the guidelines set by the U.S. Department of Health and Human Services (2008) for 60 minutes per day of moderate to vigorous physical activity, and activity levels appear to decline across age groups as children move into adolescence and beyond.[3]

As suggested by the Physical Activity Guidelines Midcourse Report, schools could provide the ideal settings, in which to make available opportunities for physical activity, since children spend the majority of their day at school (U.S. Department of Health and Human Services, 2012).[4] Potential mechanisms suggested to increase physical activity include enhanced physical education programming, activity breaks during classroom time, and active transport to and from school. Unfortunately, school-related physical activity from all sources varies widely within and across schools. For example, 44% of school administrators reported having to cut time from physical education, the arts, and recess to accommodate more time for mathematics and reading instruction since the passage of the No Child Left Behind Act of 2001.

While generally 150 minutes weekly of physical education is recommended,[5] many schools only offer physical education one to two times a week (30–40 minutes) and provide recess or other curriculum-based physical activity programs

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to make up the difference. While the physical education curriculum provides opportunity for physical activity, alone it will not enable children to meet the guideline of at least 60 minutes per day of moderate- to vigorous-intensity physical activity, regardless of how often it is offered because of the required instructional time that may not be physically active.

A focus on the effects of other physical activity options in the school environment, including recess, is therefore necessary to fill in the gaps for those implementing programs and policies to enhance or improve student health and achievement.[5] Currently, the position of the National Association for Sport and Physical Education (NASPE) is that all elementary school children should have at least one daily period of recess for at least 20 minutes. The NASPE and various other federal organizations believe recess to be a necessary educational support activity, providing an opportunity for students to develop physical, social, and academic skills.[6]

As an alternative to overall lost active time among U.S. children, researchers have explored the usefulness of exergaming, which uses active videogaming systems to increase physical activity and energy expenditure.[7–13] A recent review of studies that used off-the-shelf active videogames and quantified intensity and outcomes suggested that participation in active videogames does elicit light to moderate physical activity levels in participants.[14] Another review indicated that playing active videogames acutely increased energy expenditure in children compared to sedentary videogames; however, the energy expenditure was less than that achieved with more traditional physical activities.[15]

However, advances in technology such as those found in Xbox® Kinect® (Microsoft Corp., Redmond, WA), allow players to perform full-body natural movements without holding on to external devices. This technology provides the opportunity to enhance the intensity level of children, who play by promoting more whole-body activity as opposed to just movement of the controller.

In a study comparing the physiologic responses and energy expenditure of rest, a traditional sedentary videogame and two Kinect active videogame, *Dance Central* and *Kinect Sports Boxing*, data suggested that gameplay with Kinect active videogames produced higher heart rates, oxygen consumption, and energy expenditure.[16] In addition, the oxygen demands of *Kinect Sports Boxing* were greater than those reported in a previous study of active videogame play using *Wii Boxing*.[17] Researchers Smallwood et al. hypothesized that the absence of a game controller could explain these differences.[16]

The level of energy expenditure provided through the use of active videogames remains controversial; a recent review of studies utilizing active videogames suggested that these games led to only light- to moderate-intensity physical activity, and when done in a home setting resulted in even less energy expenditure.[18] Although activity levels and energy expenditure during active gaming may not be a sufficient alternative to participation in sports or even outside play in terms of exercise intensity, the increased energy expenditure over sedentary activities, if adhered to over a substantial period of time, could fill the energy gap observed in many children today.[19] Exergaming may also be useful as a break in extended periods of inactivity, which has been identified as a serious public health issue.

Developing physical activity habits at a young age is an important factor in limiting poor health outcomes and potentially enhancing academic performance. Given the amount of time children spend in schools and the fact that schools generally provide an equal opportunity for physical activity across the student population, they offer a setting to enhance children's physical activity levels. Active videogames have been shown to elicit greater physiological responses than observed during rest or sedentary activities in adolescents and younger children and, as such, could be used as alternative programming during recess or classroom breaks when opportunity arises.

Fogel et al. showed that exergaming enhanced physical activity in fifth grade children who were inactive during a physical education class.[12] Shayne et al. extended these results to show the effectiveness of exergaming to enhance physical activity among third grade children.[13] While both these studies examined the effect of exergaming during a physical education class, a limited number of studies have focused on using exergaming during recess. Two such studies looked at the impact of exergaming during recess on accumulated daily steps.[20,21] However, the benefits of exergaming during recess for children in terms of physiological response and energy expenditure have not been thoroughly studied.

Given the need to increase physical activity levels in children and considering the pressures on school systems to show academic growth on standardized tests, it is important to identify ways to augment physical activity opportunities for all children during school hours. Recess represents a consistent opportunity for school-based physical activity when weather permits. Exergaming could provide a viable alternative for normal recess activities for certain children or during inclement weather, or perhaps as an opportune classroom break; however, a more thorough investigation of physiological response is necessary to understand this potential. Accelerometers can estimate the total dose of moderate- to vigorous-intensity physical activity performed, providing an objective look at this option. Hence, the purpose of the current study was to assess differences in physiologic response and enjoyment in elementary school children during Xbox Kinect gameplay and regular recess activities.

Materials and Methods

Participants

Twenty-four elementary school children (mean ± SD, age = 9.17 ± 0.76 years; height = 142.82 ± 8.61 cm; mass = 41.84 ± 12.39 kg; body mass index [BMI] = 20.21 ± 4.28 kg/m^2) were randomly selected to participate in the study from a list of children, whose parents had agreed for their children to participate. Four girls and four boys were recruited from each of the third, fourth, and fifth grade classes. Recruitment was done on a ''Meet the teacher'' night. Before participation, signed informed consents were obtained from the children's parents, and children signed assent forms. Parents also completed a health status questionnaire for their child. Children with any musculoskeletal problems or any systemic health concerns (such as asthma, heart conditions, diabetes, or any neurological disease) were excluded from the study. The study was approved by the Elon University's Institutional Review Board and the local county school system.

Before recess activities were assessed, all participants completed the physical activity questionnaire for children estimating baseline physical activity on a scale of 1–5.[22] Score

on the Physical Activity Questionnaire for Children was 2.90 ± 0.74 (mean ± SD), indicating that the current study's cohort was moderately active. Using the CDC percentile curves based on age and BMI, a cutoff value of 85% or above was used to categorize 13 participants as OW and 11 participants as not overweight (NW) for this study. Fourteen children self-reported to have played Xbox Kinect before participating in the study and only two among them had played the game that was used in the current study (details below).

Sessions

At this school, each K-5 class had a scheduled 40-minute physical education class once per week. The four other days of the week, a recess was scheduled for about 20 minutes when weather permitted. The recess timings were different for third, fourth, and fifth graders. The recess was unstructured with the students having the options to play in the playground (Fig. 1, bottom half).

All the participants were evaluated in two different sessions during recess as follows: Xbox Kinect session (KINECT) and regular activity (REGULAR). The sessions were at least 1 week apart, with the order of KINECT and REGULAR sessions randomized among participants. During KINECT, children participated in an interactive videogame for ~15 minutes. All the participants played the same videogame, *Reflex Ridge on Kinect Adventures* for the study. The videogame mimicked an obstacle course race that required the participants to bend, jump, and shift from side-to-side to avoid obstacles. The game had nine different and progressively challenging courses. All the participants familiarized themselves with the game by completing the first course for practice. This familiarization course was not included for data analysis. There was an area taped off to help give visual cues to children, on where they should be in space to best perform as required by the Kinect camera (Fig. 1). From the second level onward, the children self-selected progression to the next level. Children were observed by researchers during game play without the use of a specific observation tool (Fig. 1, top half).

During REGULAR, participants were observed by researchers as they self-selected activities for the allotted recess time. Researchers did not interrupt or interfere in these activities, which generally included running and walking around the playground, playing with friends, and using monkey bars (Fig. 1, bottom half). On both occasions, a specific observation tool was not used.

Evaluations and outcome measures

Physical activity (counts/min) was objectively measured during both sessions using a wGT3X activity monitor worn around the waist (Actigraph, Pensacola, FL). Rate of energy expenditure, metabolic equivalents of task (METS), time in moderate to vigorous physical activity (minutes), percentage of sedentary, light, lifestyle, moderate, vigorous, and very vigorous activity levels were calculated from the physical activity measurement using ActiLife 6.9 software (Actigraph, Pensacola, FL). Heart rate and rate of perceived exertion (RPE) were measured just before (resting heart rate [RHR]) and after each

FIG. 1. Picture showing children participating in KINECT session (top half) and REGULAR session (bottom half) during recess. (Color images available at www.liebertonline.com/g4h)

activity session (EndHR). Heart rate was measured using heart rate monitors (Polar Electro, Inc., Lake Success, NY). RPE was measured on a scale of 1–10 with 1 being ''not tired at all'' and 10 being ''very tired.'' The level of enjoyment for each activity was measured using the Physical Activity Enjoyment Scale.[23]

Statistical analyses

After checking for normality, a paired-samples *t*-test was conducted to compare the two sessions (KINECT vs. REGULAR). As a secondary analysis, the effects of grade, gender, familiarity playing Xbox Kinect, and BMI were estimated using a separate two-way ANOVA. In each ANOVA, session (KINECT, REGULAR) was a within-subject factor. Grade (3rd, 4th, 5th), gender (boys, girls), familiarity (none, familiar), and BMI (NW, OW) were used as the respective between-subject factors. When appropriate, a Bonferroni post hoc analysis was performed to determine any significant difference between different grade levels. An α-value of 0.05 was used. The statistical analyses were performed using the SPSS 22 software (IBM Corp., Armonk, NY).

Results

On an average, the KINECT session time was 15.61 minutes and the REGULAR session was 15.33 minutes. Hence, the following measurements are not affected by the duration of the session.

KINECT session × REGULAR session

Children reported a significantly greater RPE after the KINECT session (by 19%; $P = 0.024$). The participants had significantly greater percentage of activity at sedentary ($P = 0.019$) and very vigorous levels ($P < 0.001$) and lesser percentage of moderate activity level ($P = 0.008$) during KINECT session. There was also a trend toward significance for METS, with a greater value recorded after the KINECT session compared to the REGULAR session ($P = 0.069$). The other dependent variables were similar across both the sessions (Table 1).

Since the secondary purpose of the current study was to examine the effects of grade, gender, familiarity, and BMI on the mode of play, the following sections do not focus on the significant main effects of session. The main effects of session are presented in Tables 2–4.

Grade × session

A significant interaction was found for rate of energy expenditure ($P = 0.003$), EndHR ($P = 0.031$), and METS ($P = 0.001$) when comparing grade to session. While the third and fourth graders had a greater rate of energy expenditure and METS during the KINECT session, the fifth graders had a greater rate of energy expenditure and METS during the REGULAR session. The third and fifth graders had a higher EndHR after the REGULAR session, whereas the fourth graders had a higher EndHR after the KINECT session. Although no significant main effects for grade were observed, a trend toward significance for RHR showed that the fifth graders had a higher value compared to third and fourth graders ($P = 0.058$). No significant effects were observed for any other dependent measures (Table 2).

Gender × session

No significant interactions were observed for gender and session. There were no significant gender main effects. However, there was a trend toward significance, where girls achieved a rate of energy expenditure 34% less than boys ($P = 0.058$). No significant effects were observed for any other dependent measures (Table 3).

Familiarity × session

A significant interaction was observed for EndHR, where those who were not familiar had higher EndHR after the KINECT session compared to the REGULAR session ($P = 0.041$). Conversely, those who were familiar had lower EndHRs after the KINECT session compared to the REGULAR session. There was a trend toward a significant interaction for RPE as those who were not familiar reported greater perceived exertion during the KINECT session compared to those who were familiar ($P = 0.067$). No significant effects were observed for any other dependent measures (Table 4).

Table 1. Mean (Standard Error) of Dependent Variables Comparing Both the Sessions

Dependent Variable	*KINECT*	*REGULAR*	P
EE (Kcals/min)	1.83 (0.18)	1.59 (0.21)	0.102
METS (/min)	4.21 (0.30)	3.70 (0.35)	0.069
RHR (bpm)	97.33 (3.52)	95.54 (3.38)	0.619
EndHR (bpm)	152.17 (5.57)	156.79 (6.45)	0.607
RPE	6.21 (0.44)	5.21 (0.45)	0.024[a]
PACES	3.06 (0.04)	2.96 (0.06)	0.123
Time in MVPA (min)	6.70 (3.74)	8.11 (4.65)	0.133
% of sedentary activity level	35.82 (21.16)	22.23 (17.40)	0.019[a]
% of light activity level	10.40 (3.71)	9.27 (5.53)	0.190
% of lifestyle activity level	13.63 (7.69)	16.89 (10.23)	0.110
% of moderate activity level	26.42 (19.87)	44.22 (22.04)	0.008[a]
% of vigorous activity level	6.21 (3.77)	6.27 (4.46)	0.480
% of very vigorous activity level	7.52 (5.68)	1.14 (1.02)	<0.001[a]

[a]Significant difference ($P < 0.05$).

EE, rate of energy expenditure; EndHR, heart rate at the end of a session; METS, metabolic equivalents of task; MVPA, moderate to vigorous physical activity; PACES, Physical Activity Enjoyment Scale score; RHR, resting heart rate; RPE, rate of perceived exertion.

Table 2. Mean (Standard Error) of Dependent Variables Categorized by Grade Comparing Both the Sessions

Dependent variable	*Grade*	*KINECT*	*REGULAR*	*Interaction* P-*value*	*Session main effect* P-*value*	*Grade main effect* P-*value*
EE (Kcals/min)	3rd	1.56 (0.32)	1.33 (0.34)	0.003[a]	0.043[a]	0.404
	4th	2.03 (0.32)	1.23 (0.34)			
	5th	1.90 (0.32)	2.20 (0.34)			
METS (/min)	3rd	4.09 (0.52)	3.24 (0.61)	0.001[a]	0.020[a]	0.803
	4th	4.77 (0.52)	3.40 (0.61)			
	5th	3.77 (0.52)	4.47 (0.61)			
RHR (bpm)	3rd	96.75 (5.38)	92.00 (5.82)	0.374	0.619	0.058[b]
	4th	86.50 (5.38)	91.88 (5.82)			
	5th	108.75 (5.38)	102.75 (5.82)			
EndHR (bpm)	3rd	144.00 (9.20)	158.00 (10.72)	0.031[a]	0.563	0.794
	4th	167.73 (9.20)	141.13 (10.72)			
	5th	144.88 (9.20)	171.25 (10.72)			
RPE	3rd	6.50 (0.79)	5.38 (0.81)	0.287	0.024[a]	0.925
	4th	6.50 (0.79)	4.75 (0.81)			
	5th	5.63 (0.79)	5.50 (0.81)			
PACES	3rd	2.99 (0.08)	2.84 (0.10)	0.605	0.133	0.147
	4th	3.09 (0.08)	2.93 (0.10)			
	5th	3.12 (0.08)	3.11 (0.10)			

[a]Significant difference ($P \leq 0.05$).
[b]Trend toward significance ($0.051 \leq P \leq 0.070$).

BMI × session

No significant interactions were observed between BMI and session. There was also a trend toward a significant BMI main effect for RPE ($P = 0.062$). Participants in the OW group had a 23% higher RPE than those in the NW group. No significant effects were observed for any other dependent measures (Table 4).

Discussion

The purpose of the current study was to assess differences in physiologic response and enjoyment in elementary school children during Xbox Kinect game play and regular recess activities. Overall, we found that playing Kinect was similar to participating in a self-selected regular activity during recess. Since the duration of the sessions was not significantly different, the findings were not influenced by this factor.

However, while playing Kinect, children perceived that they worked harder, although with similar energy expenditure and perception of enjoyment and slightly more METS than while on the playground. These differences combined with those seen in the sedentary, moderate, and very vigorous activity levels between the KINECT and REGULAR sessions could be attributed to the nature of the activity within each session. For example, exergaming during KINECT session was more structured requiring fast movements during the game and pauses between each levels, whereas the regular recess activities were more unstructured and required moderate to vigorous physical activities. Also, participants played the Kinect game by themselves, whereas during regular recess activity they

Table 3. Mean (Standard Error) of Dependent Variables Categorized by Gender Comparing Both the Sessions

Dependent variable	*Gender*	*KINECT*	*REGULAR*	*Interaction* P-*value*	*Session main effect* P-*value*	*Gender main effect* P-*value*
EE (Kcals/min)	Girls	1.48 (0.24)	1.24 (0.29)	0.993	0.110	0.058[a]
	Boys	2.18 (0.24)	1.94 (0.29)			
METS	Girls	4.05 (0.43)	3.32 (0.43)	0.415	0.072	0.381
	Boys	4.37 (0.50)	4.09 (0.50)			
RHR	Girls	94.17 (5.00)	93.25 (4.84)	0.812	0.626	0.369
	Boys	100.50 (5.00)	97.83 (4.84)			
EndHR	Girls	146.00 (7.83)	154.75 (9.30)	0.652	0.613	0.325
	Boys	158.33 (7.83)	158.83 (9.30)			
RPE	Girls	6.00 (0.64)	5.33 (0.66)	0.436	0.026[b]	0.919
	Boys	6.42 (0.64)	5.08 (0.66)			
PACES	Girls	3.08 (0.06)	2.98 (0.09)	0.933	0.132	0.597
	Boys	3.04 (0.06)	2.93 (0.09)			

[a]Trend toward significance ($0.051 \leq P \leq 0.070$).
[b]Significant difference ($P \leq 0.05$).

Table 4. Mean (Standard Error) of Dependent Variables Categorized by Familiarity and Body Mass Index Comparing Both the Sessions

Factor	*Levels*	*Dependent measure*	*KINECT*	*REGULAR*	*Interaction* P-*value*	*Session main effect* P-*value*	*Familiarity/BMI effect* P-*value*
Familiarity	None	EE (Kcals/min)	2.14 (0.28)	1.79 (0.33)	0.514	0.091	0.245
	Familiar	EE (Kcals/min)	1.61 (0.23)	1.45 (0.28)			
	None	METS (/min)	4.71 (0.46)	3.79 (0.56)	0.201	0.045[a]	0.422
	Familiar	METS (/min)	3.86 (0.38)	3.64 (0.47)			
	None	RHR (bpm)	94.10 (5.51)	91.20 (5.22)	0.799	0.601	0.290
	Familiar	RHR (bpm)	99.64 (4.65)	98.64 (4.41)			
	None	EndHR (bpm)	171.20 (7.03)	154.70 (10.19)	0.041[a]	0.849	0.078
	Familiar	EndHR (bpm)	138.57 (5.95)	158.29 (8.62)			
	None	RPE	6.80 (0.68)	4.90 (0.71)	0.067[b]	0.010[a]	0.770
	Familiar	RPE	5.79 (0.58)	5.43 (0.60)			
	None	PACES	3.10 (0.07)	2.99 (0.10)	0.974	0.136	0.480
	Familiar	PACES	3.04 (0.06)	2.93 (0.08)			
BMI	NW	EE (Kcals/min)	1.95 (0.27)	1.70 (0.32)	0.956	0.110	0.590
	OW	EE (Kcals/min)	1.73 (0.25)	1.50 (0.29)			
	NW	METS (/min)	3.83 (0.44)	3.13 (0.51)	0.516	0.067[b]	0.146
	OW	METS (/min)	4.53 (0.40)	4.19 (0.47)			
	NW	RHR (bpm)	98.27 (5.31)	93.73 (5.08)	0.488	0.584	0.895
	OW	RHR (bpm)	96.54 (4.89)	97.08 (4.68)			
	NW	EndHR (bpm)	154.73 (8.37)	142.09 (8.76)	0.072	0.700	0.176
	OW	EndHR (bpm)	150.00 (7.70)	169.23 (8.05)			
	NW	RPE	5.36 (0.62)	4.45 (0.65)	0.846	0.030[a]	0.062[b]
	OW	RPE	6.92 (0.57)	5.85 (0.60)			
	NW	PACES	3.02 (0.07)	3.04 (0.09)	0.086	0.143	0.668
	OW	PACES	3.10 (0.06)	2.89 (0.08)			

[a]Significant difference ($P \leq 0.05$).
[b]Trend toward significance ($0.051 \leq P \leq 0.070$).
BMI, body mass index; NW, not overweight; OW, overweight.

predominately participated in combined activities with their peers. The participants had the opportunity to self-select activities and the intensity of the activity during regular recess activity. In contrast, their activity and intensity was guided by Kinect, thereby possibly affecting their RPE.

Future research should focus on implementing the design of the current study using a multiplayer mode to mediate the variability that may result from direct observation of solo Kinect play. Using the same game on Xbox Kinect, O'Donovan et al. showed that healthy young adults expended more energy and achieved higher heart rates in multiplayer mode compared to single-player mode.[24] Previous literature also indicates that multiplayer mode is more exciting and generates higher galvanic skin response.[25] The type of game used in the single- or multiplayer mode may also affect the results. For example, games such as Reflex Ridge challenge the participant to complete the course individually, but games such as Kinect Sports require interaction with an opponent, potentially enhancing the overall experience of the play. In addition, allowing the child to self-select the game to play could influence the level of engagement and intensity in the physical activity, as well as the enjoyment, affecting the utility of Kinect as resource for increasing physical activity.

Differences in RPE could also have been due to the temperature and humidity levels. The average outside temperature on data collection days was 70°F/21.1°C. Although it was impractical to control the temperature differences between the sessions, such a difference was expected as part of the ecological validity of the study. Previous studies on the effect of temperature on physical activity suggested that children and adolescents spend greater amount of time and play with more intensity on longer and warmer days.[26,27] However, the amount of vigorous physical activity has also been shown to be lesser at higher temperatures among elementary school children during recess.[28] Results of the current study are in line with the suggestion from Ridgers et al. that an option of exergaming could help provide a more vigorous activity when faced with unmodifiable factors such as temperature, which could limit the physical activity opportunities during recess.[28]

In this study, average METS for both sessions fell within the range of 3.0–6.0, which has been considered as moderate to vigorous physical activity in adults.[2,16] Interestingly, METS during KINECT session were more than four, which is considered as moderate physical activity for children, whereas the METS value during regular recess was less than 4. Moreover, the mean heart rates of 152 and 156 bpm observed during KINECT and REGULAR, respectively, were equivalent to 72% and 74% of estimated maximum heart rate (calculated as 220 average age of participants), offering support that children achieved at least a moderate level of physical activity. This is consistent with responses observed in studies with adolescents.[9,16]

Fifth graders expended more energy and METS during regular recess compared to third and fourth graders who expended more energy playing Kinect. This could be due to choice of activity during recess. The recess times for all the grades were at different times for the school, in which the study was conducted. While third and fourth graders mostly participated in activities, such as using monkey bars, fifth

graders spent their time running on the playground. It is unclear why the fourth graders had higher ending heart rate after playing Kinect compared to third and fifth graders. Although results did not show significant differences, the values for energy expenditure, METS, and RPE followed a similar pattern as that of EndHR. Perhaps the fourth graders felt more engaged while playing the Kinect game compared to the other two groups. Perception of engagement in a virtual environment could potentially be evaluated using a questionnaire.

There was a trend that OW children perceived they worked harder than NW children in both KINECT and REGULAR sessions, although there were no significant differences in the physiological variables measured. This may be explained, in part, by the design of the study. While Kinect was played individually under direct observation, recess activities were often performed in groups, enabling OW children to moderate their intensity of play without the perception that they were being directly observed. During KINECT, direct observation and the need to meet standards for silver or gold medals before advancing required more focus on the part of participants, and perhaps enhanced the perception of exertion on the part of the OW participants.

It is possible that the use of a tool, such as the Scheme for Observing Activity Level, which enables quantification of type and frequency of activity in children, could be valuable to further determine actual level of participation in activities of recess and Kinect play.[29] Other tools, such as System for Observing Children's Activity and Relationships during Play[30] and Observation Tool for Active Gaming and Movement,[31] could also be used.

In the current study, rate of energy expenditure seemed to be less in girls compared to boys. While a greater sample size is required to confirm, these findings are consistent to what is generally found in the literature. Also, although children were given a practice run before the actual data collection started, familiarity with Kinect was observed to impact physiologic response. It appears that more experience with the gaming system enables participants to mediate full-body movement and still advance to the next level of the game. This suggests that with increasing familiarity with the gaming system, the efficacy of Kinect as an appropriate mode of physical activity may decline. Hence, appropriately using a variety of games might enhance the effectiveness of using exergaming as a viable option to enhance physical activity in children.

Regardless, the energy expenditure is greater than what a child is likely to achieve with a sedentary activity, as suggested by Smallwood et al., supporting the use of appropriate active videogames in addressing issues of inactivity and potential weight gain in children.[16] Results of the current study are also in line with those published earlier supporting the use of exergaming for elementary-aged school children as a means for enhancing physical activity.[12,13] Duncan and Staples reported that there was an initial increase in the amount of physical activity (measured using number of steps taken in a day) among those children who played Nintendo® Wii™ active videogames during school recess compared to those who did not.[20] However, after 6 weeks, the total number of steps per day were lesser for those who used exergaming during recess.[20] These results combined with the results of the current study suggest that exergaming might be a good viable supplement to free play, but cannot be a standalone mode of physical activity during school recess.

The sample size used in the current pilot study, although small, is representative of the sample sizes found in the literature examining the effects of active videogame use for physical activity purposes in children and adolescents.[7–9,12,13,16,17,19–21] The current study has high ecological validity, given that it was conducted in an actual school setting during recess. In addition, the current study was strengthened by the use of a crossover design, which controlled for intrasubject variability. However, a larger sample size is warranted to confirm these results.

Conclusion

Overall, results of this pilot study suggest that playing Kinect can elicit similar physiologic responses, perceived exertion, and enjoyment from elementary school children as normal activities of recess. Xbox Kinect could be used as a relatively cost-effective supplement to recess activity, either when outside play is limited due to environmental conditions, or before or after school. Playing Kinect may also be a viable option when the opportunity to play with peers is unavailable and the child needs to find a way to be active alone.

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References

1. Ogden CL, Carroll MD, Curtin LR, et al. Prevalence of high body mass index in US children and adolescents, 2007–2008. JAMA 2010; 303:242–249.
2. Physical Activity Guidelines. health.gov. http://health.gov/paguidelines (accessed April 14, 2016).
3. Troiano RP, Berrigan D, Dodd KW, et al. Physical activity in the United States measured by Accelerometer. Med Sci Sports Exerc 2008; 40:181–188.
4. Physical Activity Guidelines for Americans Midcourse Report Subcommittee of the President's Council on Fitness, Sports & Nutrition. *Physical Activity Guidelines for Americans Midcourse Report: Strategies to Increase Physical Activity Among Youth*. Washington, DC: U.S. Department of Health and Human Services, 2012. http://health.gov/paguidelines/midcourse/pag-mid-course-report-final.pdf (accessed April 14, 2016).
5. Kohl HW, Cook HD. *Educating the Student Body: Taking Physical Activity and Physical Education to School*. In Committee on Physical Activity and Physical Education in the School Environment; Food and Nutrition Board; Institute of Medicine. Washington, DC: The National Academies Press; 2013.

6. National Association for Sport and Physical Education. *Recess for Elementary School Students*. [Position Statement] 2006.
7. Maddison R, Mhurchu CN, Jull A, et al. Energy expended playing video console games: An opportunity to increase children's physical activity? Pediatr Exerc Sci 2007; 19: 334–343.
8. Mellecker RR, McManus AM. Energy expenditure and cardiovascular responses to seated and active gaming in children. Arch Pediatr Adolesc Med 2008; 162:886–891.
9. Graf DL, Pratt LV, Hester CN, Short KR. Playing active video games increases energy expenditure in children. Pediatrics 2009; 124:534–540.
10. Dixon R, Maddison R, Ni Mhurchu C, et al. Parents' and children's perceptions of active video games: A focus group study. J Child Health Care Prof Work Child Hosp Community 2010; 14:189–199.
11. Biddiss E, Irwin J. Active video games to promote physical activity in children and youth: A systematic review. Arch Pediatr Adolesc Med 2010; 164:664–672.
12. Fogel VA, Miltenberger RG, Graves R, Koehler S. The effects of exergaming on physical activity among inactive children in a physical education classroom. J Appl Behav Anal 2010; 43:591–600.
13. Shayne RK, Fogel VA, Miltenberger RG, Koehler S. The effects of exergaming on physical activity in a third-grade physical education class. J Appl Behav Anal 2012; 45:211–215.
14. Peng W, Crouse JC, Lin J-H. Using active video games for physical activity promotion: A systematic review of the current state of research. Health Educ Behav Off Publ Soc Public Health Educ 2013; 40:171–192.
15. LeBlanc AG, Chaput J-P, McFarlane A, et al. Active video games and health indicators in children and youth: A systematic review. PLoS One 2013; 8:e65351.
16. Smallwood SR, Morris MM, Fallows SJ, Buckley JP. Physiologic responses and energy expenditure of kinect active video game play in schoolchildren. Arch Pediatr Adolesc Med 2012; 166:1005–1009.
17. Graves LEF, Ridgers ND, Stratton G. The contribution of upper limb and total body movement to adolescents' energy expenditure whilst playing Nintendo Wii. Eur J Appl Physiol 2008; 104:617–623.
18. Liang Y, Lau PWC. Effects of active videogames on physical activity and related outcomes among healthy children: A systematic review. Games Health J 2014; 3:122–144.
19. Graves L, Stratton G, Ridgers ND, Cable NT. Comparison of energy expenditure in adolescents when playing new generation and sedentary computer games: Cross sectional study. BMJ 2007; 335:1282–1284.
20. Duncan MJ, Staples V. The impact of a school-based active video game play intervention on children's physical activity during recess. Hum Mov 2010; 11:95–99.
21. McDougall J, Duncan MJ. Children, videogames and physical activity: An exploratory Study. Int J Disability Hum Dev 2008; 7: 89–94.
22. Crocker PR, Bailey DA, Faulkner RA, et al. Measuring general levels of physical activity: Preliminary evidence for the Physical Activity Questionnaire for Older Children. Med Sci Sports Exerc 1997; 29:1344–1349.
23. Moore JB, Yin Z, Hanes J, et al. Measuring enjoyment of physical activity in children: Validation of the Physical Activity Enjoyment Scale. J Appl Sport Psychol 2009; 21(S1):S116–S129.
24. O'Donovan C, Hirsch E, Holohan E, et al. Energy expended playing Xbox Kinect™ and Wii™ games: A preliminary study comparing single and multiplayer modes. Physiotherapy 2012; 98:224–229.
25. Mandryk RL, Inkpen KM, Calvert TW. Using psychophysiological techniques to measure user experience with entertainment technologies. Behav Inf Technol 2006; 25:141–158.
26. Goodman A, Paskins J, Mackett R. Day length and weather effects on children's physical activity and participation in play, sports, and active travel. J Phys Act Health 2012; 9:1105–1116.
27. Bélanger M, Gray-Donald K, O'Loughlin J, et al. Influence of weather conditions and season on physical activity in adolescents. Ann Epidemiol 2009; 19:180–186.
28. Ridgers ND, Fairclough SJ, Stratton G. Variables associated with children's physical activity levels during recess: The A-CLASS project. Int J Behav Nutr Phys Act 2010; 7:74.
29. Eaton WO, Enns LR, Presse MC. Scheme for observing activity level: reliability and convergent validity. J Psychoeduc Assess 1987; 5:273–280.
30. Ridgers ND, Stratton G, McKenzie TL. Reliability and validity of the System for observing Children's Activity and Relationships during Play (SOCARP). J Phys Act Health 2010; 7:17–25.
31. Rosa RL, Ridgers ND, Barnett LM. Development and use of an observation tool for active gaming and movement (OTAGM) to measure children's movement skill components during active video game play. Percept Mot Skills 2013; 117:935–949.

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